

Coremoni-WE: Individual Core Training Monitoring and Support System Using an IMU at the Waist and the Ear

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ABSTRACT

Due to the influence of the new coronavirus, many people are interrupting fitness clubs and exercise/sports performed by multiple people. Under these circumstances, "core training," which can be easily performed indoors by individuals, attracts attention as an exercise to improve health. However, it is not easy to recognize whether the posture during the training is correct or not, which may significantly reduce the effect of the exercise. To tackle these issues, we have been developing "Coremoni-WE," a core training monitoring and support system that combines wearables and earables to judge posture.

CCS CONCEPTS

• **Computer systems organization** → **Embedded systems**; **Redundancy**; **Robotics**; • **Networks** → **Network reliability**.

KEYWORDS

wearable device, sports, fitness monitoring

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1 INTRODUCTION

In recent years, people have been spending more and more time at home to prevent the spread of new coronaviruses. In a survey on health[2], more than 30% of the respondents answered that they became more active after the coronavirus outbreak, and "core

training," which can be easily performed indoors by individuals, is attracting attention. However, there is a concern that the effects of trunk training performed by individuals may be reduced because the trunk is often not used correctly compared to the activity conducted under an instructor. This study proposes a support system for core training using wearable and earable IMUs to help individuals perform core training correctly.

2 RELATED WORKS

Barbado et al. studied about the reliability of smartphone accelerometers to quantify the intensity of core training [1]. From the five types of core training performed by 23 participants, they obtained moderate-to-high reliability scores for pelvic acceleration and concluded that smartphone accelerometers is useful to identify the individuals' core training status and to improve the training programs. However, this study is not mentioned about the correct postures and user feedbacks.

While many researchers and developers have been developing applications based on smartphones and smartwatches, Kawsar et al. [3] proposed and developed a new wearable platform called "eSense". The eSense platform consists of a pair of wireless earbuds augmented with kinetic, audio, and proximity sensing [4]. The left earbud has a six-axis IMU with an accelerometer, a gyroscope, and a Bluetooth Low Energy (BLE) interface to stream sensor data to a paired smartphone. Both earbuds are also equipped with microphones to record external sounds.

Prakash et al. introduced the advantages of eSense in counting the number of steps of walking [5]. While the head movement can still pollute this bouncing signal, they proposed a method to alleviate the noise from head movement. Results show 95% step count accuracy even in the most challenging test case—very slow walk—where smartphone and wrist-band-type systems falter. Importantly, their system, which is named "STEAR" (STEP counting from EARables) is robust to changes in walking patterns and scales well across different users. Additionally, they demonstrated how STEAR also brings opportunities for practical jump analysis, often crucial for exercises and injury-related rehabilitation.

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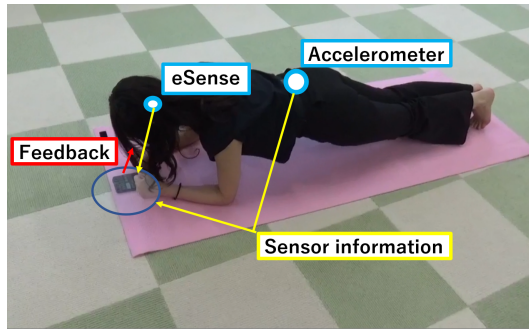


Figure 1: configuration of CoreMoni-WE

3 PROPOSED SYSTEM

3.1 CoreMoni Overview

CoreMoni is a core training monitoring and support system that focuses on the user's "posture" and "trunk blurring" in multiple core training exercises. It guides to improve the "posture" and "trunk blurring." In this study, we selected a method to determine "appropriate posture" using wearable and earable sensors (see usage image in Fig. 1). CoreMoni collects the acceleration and angular velocity values of the trunk and head from a wearable acceleration sensor attached to the user's waist (movesense¹) and an earable acceleration sensor attached to the head (eSense²), and transmits the data to a mobile information terminal (smartphone or tablet PC). The transmitted data is converted to variance and angle values and used for feedback on the mobile terminal. Two types of feedback are provided: switching images for "posture" and sound effects for "blur of the trunk" during the trunk training.

3.2 Examination of the number of accelerometers installed and their positions

We conducted a requirement extraction experiment to determine the number and position of accelerometers in the CoreMoni-WE implementation. One university student in his 20's was asked to perform front plank, which is the most basic core training exercise. We attached accelerometers to the subject's head, waist, and ankles and asked him to perform front plank for one minute according to the following procedure.

- 0 - 15 sec: correct posture
- 15 - 30 sec: low back posture
- 30 - 45 sec: high back posture
- 45 - 60 sec: continuously changing the posture

We recorded the acceleration at each sensor position during the elapsed time of 1 minute of core training. After the experiment, the results were plotted for analysis. Acceleration of the sensors attached to the head and the ankle does not change much along any axis. On the other hand, the acceleration of the sensor attached to the waist showed some change, but the difference was significant only in the y-axis. We can assume that the y-axis of acceleration at the waist is useful to detect "trunk blur." Since two points are

¹movesense, SUUNTO, <https://www.movesense.com/>

²eSense, Nokia Bell Labs, Cambridge, <https://www.esense.io/>

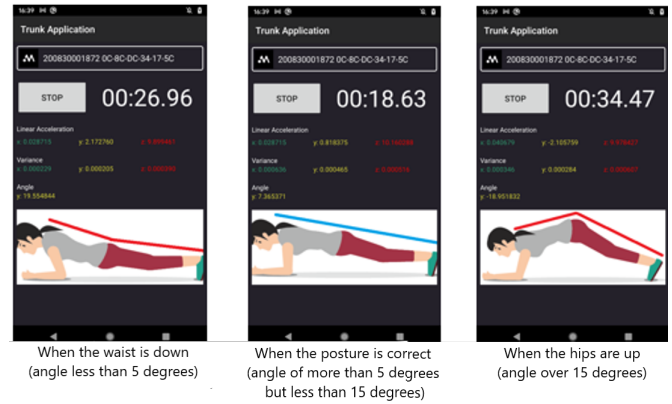


Figure 2: CoreMoni-WE's visual feedback for front plank

preferable to estimate trunk posture, and no difference was observed between acceleration at the ankle and the head, in addition to the wearable sensor at the waist, we selected the earable sensor at the head, assuming it is more likely to be used than a wearable sensor at the ankle. A feedback application based only on the waist attached wearable sensor output as been developed, using thresholds on y-axis acceleration for evaluating "posture" (see Fig. 2).

4 FUTURE WORKS

This paper proposed supporting individual core training with a system using accelerometers attached to the waist and ear. Our first application prototype can judge front plank posture using only waist attached wearable sensor. We are designing new algorithms combining waist attached wearable sensor and head attached earable sensor to discriminate and support multiple core training exercises. In this study, we focused only on acceleration among the data that can be collected from wearable and earable sensors. Still, in the future, it may be possible to perform training more following the user's health condition by collecting biological signals such as heart rate.

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